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LEARNING AIDS FOR THIS CHAPTER ON OUR WEBSITE (www.mhhe.com/hillier)

Solved Examples:

Examples for Chapter 7

A Demonstration Example in OR Tutor:

Sensitivity Analysis

Interactive Procedures in IOR Tutorial:

Interactive Graphical Method
Enter or Revise a General Linear Programming Model
Solve Interactively by the Simplex Method
Sensitivity Analysis

Automatic Procedures in IOR Tutorial:

Solve Automatically by the Simplex Method
Graphical Method and Sensitivity Analysis

Excel Add-In:

Analytic Solver Platform for Education (ASPE)

Files (Chapter 3) for Solving the Wyndor Example:

Excel Files
LINGO/LINDO File
MPL/Solvers File

Glossary for Chapter 7

See Appendix 1 for documentation of the software.

PROBLEMS

The symbols to the left of some of the problems (or their parts) have the following meaning:

D: The demonstration example just listed may be helpful.
I: We suggest that you use the corresponding interactive procedure just listed (the printout records your work).
C: Use the computer with any of the software options available to you (or as instructed by your instructor) to solve the problem automatically.
E*: Use Excel, perhaps including the ASPE add-in.

An asterisk on the problem number indicates that at least a partial answer is given in the back of the book.

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7.1-1.* Consider the following problem.

$$\text{Maximize} \qquad Z = 3x_1 + x_2 + 4x_3,$$

subject to

$$6x_1 + 3x_2 + 5x_3 \le 25$$
$$3x_1 + 4x_2 + 5x_3 \le 20$$

and

$$x_1 \ge 0, \qquad x_2 \ge 0, \qquad x_3 \ge 0.$$

The corresponding final set of equations yielding the optimal solution is

$$\begin{array}{lrrrrrl} (0) & Z & +\,2x_2 & & +\,\frac{1}{5}x_4 & +\,\frac{3}{5}x_5 & = 17 \\ (1) & & x_1 & -\,\frac{1}{3}x_2 & +\,\frac{1}{3}x_4 & -\,\frac{1}{3}x_5 & = \frac{5}{3} \\ (2) & & & x_2 + x_3 & -\,\frac{1}{5}x_4 & +\,\frac{2}{5}x_5 & = 3. \end{array}$$

(a) Identify the optimal solution from this set of equations.
(b) Construct the dual problem.
I **(c)** Identify the optimal solution for the dual problem from the final set of equations. Verify this solution by solving the dual problem graphically.
(d) Suppose that the original problem is changed to

$$\text{Maximize} \qquad Z = 3x_1 + 3x_2 + 4x_3,$$

subject to

$$6x_1 + 2x_2 + 5x_3 \le 25$$
$$3x_1 + 3x_2 + 5x_3 \le 20$$

and

$$x_1 \ge 0, \qquad x_2 \ge 0, \qquad x_3 \ge 0.$$

Use duality theory to determine whether the previous optimal solution is still optimal.
(e) Use the fundamental insight presented in Sec. 5.3 to identify the new coefficients of x_2 in the final set of equations after it has been adjusted for the changes in the original problem given in part (*d*).
(f) Now suppose that the only change in the original problem is that a new variable x_{new} has been introduced into the model as follows:

$$\text{Maximize} \qquad Z = 3x_1 + x_2 + 4x_3 + 2x_{\text{new}},$$

subject to

$$6x_1 + 3x_2 + 5x_3 + 3x_{\text{new}} \le 25$$
$$3x_1 + 4x_2 + 5x_3 + 2x_{\text{new}} \le 20$$

and

$$x_1 \ge 0, \qquad x_2 \ge 0, \qquad x_3 \ge 0, \qquad x_{\text{new}} \ge 0.$$

Use duality theory to determine whether the previous optimal solution, along with $x_{\text{new}} = 0$, is still optimal.
(g) Use the fundamental insight presented in Sec. 5.3 to identify the coefficients of x_{new} as a nonbasic variable in the final set of equations resulting from the introduction of x_{new} into the original model as shown in part (*f*).

D,I **7.1-2.** Reconsider the model of Prob. 7.1-1. You are now to conduct sensitivity analysis by *independently* investigating each of the following six changes in the original model. For each change, use the sensitivity analysis procedure to revise the given final set of equations (in tableau form) and convert it to proper form from Gaussian elimination. Then test this solution for feasibility and for optimality. (Do not reoptimize.)
(a) Change the right-hand side of constraint 1 to $b_1 = 10$.
(b) Change the right-hand side of constraint 2 to $b_2 = 10$.
(c) Change the coefficient of x_2 in the objective function to $c_2 = 3$.
(d) Change the coefficient of x_3 in the objective function to $c_3 = 2$.
(e) Change the coefficient of x_2 in constraint 2 to $a_{22} = 2$.
(f) Change the coefficient of x_1 in constraint 1 to $a_{11} = 8$.

D,I **7.1-3.** Consider the following problem.

$$\text{Minimize} \qquad W = 5y_1 + 4y_2,$$

subject to

$$4y_1 + 3y_2 \ge 4$$
$$2y_1 + y_2 \ge 3$$
$$y_1 + 2y_2 \ge 1$$
$$y_1 + y_2 \ge 2$$

and

$$y_1 \ge 0, \qquad y_2 \ge 0.$$

Because this primal problem has more functional constraints than variables, suppose that the simplex method has been applied directly to its dual problem. If we let x_5 and x_6 denote the slack variables for this dual problem, the resulting final simplex tableau is

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	x_6	Right Side
Z	(0)	1	3	0	2	0	1	1	9
x_2	(1)	0	1	1	−1	0	1	−1	1
x_4	(2)	0	2	0	3	1	−1	2	3

For each of the following independent changes in the original primal model, you now are to conduct sensitivity analysis by directly investigating the effect on the dual problem and then inferring the complementary effect on the primal problem. For each change, apply the procedure for sensitivity analysis summarized at the end of Sec. 7.1 to the dual problem (do *not* reoptimize), and then give your conclusions as to whether the current basic solution for the primal problem still is feasible and whether it still is optimal. Then check your conclusions by a direct graphical analysis of the primal problem.

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(a) Change the objective function to $W = 3y_1 + 5y_2$.
(b) Change the right-hand sides of the functional constraints to 3, 5, 2, and 3, respectively.
(c) Change the first constraint to $2y_1 + 4y_2 \geq 7$.
(d) Change the second constraint to $5y_1 + 2y_2 \geq 10$.

7.2-1. Read the referenced article that fully describes the OR study summarized in the application vignette presented in Sec. 7.2. Briefly describe how sensitivity analysis was applied in this study. Then list the various financial and nonfinancial benefits that resulted from the study.

D,I **7.2-2.*** Consider the following problem.

$$\text{Maximize} \qquad Z = -5x_1 + 5x_2 + 13x_3,$$

subject to

$$\begin{aligned} -x_1 + x_2 + 3x_3 &\leq 20 \\ 12x_1 + 4x_2 + 10x_3 &\leq 90 \end{aligned}$$

and

$$x_j \geq 0 \qquad (j = 1, 2, 3).$$

If we let x_4 and x_5 be the slack variables for the respective constraints, the simplex method yields the following final set of equations:

$$\begin{array}{lrcl} (0) & Z + 2x_3 + 5x_4 & = & 100 \\ (1) & -x_1 + x_2 + 3x_3 + x_4 & = & 20 \\ (2) & 16x_1 - 2x_3 - 4x_4 + x_5 & = & 10. \end{array}$$

Now you are to conduct sensitivity analysis by *independently* investigating each of the following nine changes in the original model. For each change, use the sensitivity analysis procedure to revise this set of equations (in tableau form) and convert it to proper form from Gaussian elimination for identifying and evaluating the current basic solution. Then test this solution for feasibility and for optimality. (Do not reoptimize.)

(a) Change the right-hand side of constraint 1 to

$$b_1 = 30.$$

(b) Change the right-hand side of constraint 2 to

$$b_2 = 70.$$

(c) Change the right-hand sides to

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 100 \end{bmatrix}.$$

(d) Change the coefficient of x_3 in the objective function to

$$c_3 = 8.$$

(e) Change the coefficients of x_1 to

$$\begin{bmatrix} c_1 \\ a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix}.$$

(f) Change the coefficients of x_2 to

$$\begin{bmatrix} c_2 \\ a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 5 \end{bmatrix}.$$

(g) Introduce a new variable x_6 with coefficients

$$\begin{bmatrix} c_6 \\ a_{16} \\ a_{26} \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 5 \end{bmatrix}.$$

(h) Introduce a new constraint $2x_1 + 3x_2 + 5x_3 \leq 50$. (Denote its slack variable by x_6.)
(i) Change constraint 2 to

$$10x_1 + 5x_2 + 10x_3 \leq 100.$$

7.2-3.* Reconsider the model of Prob. 7.2-2. Suppose that the right-hand sides of the functional constraints are changed to

$$20 + 2\theta \qquad \text{(for constraint 1)}$$

and

$$90 - \theta \qquad \text{(for constraint 2)},$$

where θ can be assigned any positive or negative values.

Express the basic solution (and Z) corresponding to the original optimal solution as a function of θ. Determine the lower and upper bounds on θ before this solution would become infeasible.

D,I **7.2-4.** Consider the following problem.

$$\text{Maximize} \qquad Z = 2x_1 + 7x_2 - 3x_3,$$

subject to

$$\begin{aligned} x_1 + 3x_2 + 4x_3 &\leq 30 \\ x_1 + 4x_2 - x_3 &\leq 10 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

By letting x_4 and x_5 be the slack variables for the respective constraints, the simplex method yields the following final set of equations:

$$\begin{array}{lrcl} (0) & Z + x_2 + x_3 + 2x_5 & = & 20 \\ (1) & -x_2 + 5x_3 + x_4 - x_5 & = & 20 \\ (2) & x_1 + 4x_2 - x_3 + x_5 & = & 10. \end{array}$$

Now you are to conduct sensitivity analysis by *independently* investigating each of the following seven changes in the original model. For each change, use the sensitivity analysis procedure to revise this set of equations (in tableau form) and convert it to proper form from Gaussian elimination for identifying and evaluating the current basic solution. Then test this solution for feasibility and for optimality. If either test fails, reoptimize to find a new optimal solution.

(a) Change the right-hand sides to

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$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 30 \end{bmatrix}.$$

(b) Change the coefficients of x_3 to

$$\begin{bmatrix} c_3 \\ a_{13} \\ a_{23} \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix}.$$

(c) Change the coefficients of x_1 to

$$\begin{bmatrix} c_1 \\ a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 2 \end{bmatrix}.$$

(d) Introduce a new variable x_6 with coefficients

$$\begin{bmatrix} c_6 \\ a_{16} \\ a_{26} \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}.$$

(e) Change the objective function to $Z = x_1 + 5x_2 - 2x_3$.
(f) Introduce a new constraint $3x_1 + 2x_2 + 3x_3 \leq 25$.
(g) Change constraint 2 to $x_1 + 2x_2 + 2x_3 \leq 35$.

7.2-5. Reconsider the model of Prob. 7.2-4. Suppose that the right-hand sides of the functional constraints are changed to

$$30 + 3\theta \qquad \text{(for constraint 1)}$$

and

$$10 - \theta \qquad \text{(for constraint 2)},$$

where θ can be assigned any positive or negative values.

Express the basic solution (and Z) corresponding to the original optimal solution as a function of θ. Determine the lower and upper bounds on θ before this solution would become infeasible.

D,I **7.2-6.** Consider the following problem.

$$\text{Maximize} \qquad Z = 2x_1 - x_2 + x_3,$$

subject to

$$\begin{aligned} 3x_1 - 2x_2 + 2x_3 &\leq 15 \\ -x_1 + x_2 + x_3 &\leq 3 \\ x_1 - x_2 + x_3 &\leq 4 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

If we let x_4, x_5, and x_6 be the slack variables for the respective constraints, the simplex method yields the following final set of equations:

$$\begin{array}{lrrrrrrl} (0) & Z & & +2x_3 & +x_4 & +x_5 & & = 18 \\ (1) & & x_2 & +5x_3 & +x_4 & +3x_5 & & = 24 \\ (2) & & & 2x_3 & & +x_5 & +x_6 & = 7 \\ (3) & x_1 & & +4x_3 & +x_4 & +2x_5 & & = 21. \end{array}$$

Now you are to conduct sensitivity analysis by *independently* investigating each of the following eight changes in the original model. For each change, use the sensitivity analysis procedure to revise this set of equations (in tableau form) and convert it to proper form from Gaussian elimination for identifying and evaluating the current basic solution. Then test this solution for feasibility and for optimality. If either test fails, reoptimize to find a new optimal solution.

(a) Change the right-hand sides to

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ 2 \end{bmatrix}.$$

(b) Change the coefficient of x_3 in the objective function to $c_3 = 2$.
(c) Change the coefficient of x_1 in the objective function to $c_1 = 3$.
(d) Change the coefficients of x_3 to

$$\begin{bmatrix} c_3 \\ a_{13} \\ a_{23} \\ a_{33} \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}.$$

(e) Change the coefficients of x_1 and x_2 to

$$\begin{bmatrix} c_1 \\ a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \\ 3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} c_2 \\ a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ 3 \\ 2 \end{bmatrix},$$

respectively.
(f) Change the objective function to $Z = 5x_1 + x_2 + 3x_3$.
(g) Change constraint 1 to $2x_1 - x_2 + 4x_3 \leq 12$.
(h) Introduce a new constraint $2x_1 + x_2 + 3x_3 \leq 60$.

C **7.2-7** Consider the Distribution Unlimited Co. problem presented in Sec. 3.4 and summarized in Fig. 3.13.

Although Fig. 3.13 gives estimated unit costs for shipping through the various shipping lanes, there actually is some uncertainty about what these unit costs will turn out to be. Therefore, before adopting the optimal solution given at the end of Sec. 3.4, management wants additional information about the effect of inaccuracies in estimating these unit costs.

Use a computer package based on the simplex method to generate sensitivity analysis information preparatory to addressing the following questions.

(a) Which of the unit shipping costs given in Fig. 3.13 has the smallest margin for error without invalidating the optimal solution given in Sec. 3.4? Where should the greatest effort be placed in estimating the unit shipping costs?
(b) What is the allowable range for each of the unit shipping costs?
(c) How should these allowable ranges be interpreted to management?
(d) If the estimates change for more than one of the unit shipping costs, how can you use the generated sensitivity analysis information to determine whether the optimal solution might change?

7.2-8. Consider the following problem.

$$\text{Maximize} \qquad Z = c_1x_1 + c_2x_2,$$

subject to

$$\begin{aligned} 2x_1 - x_2 &\leq b_1 \\ x_1 - x_2 &\leq b_2 \end{aligned}$$

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and

$$x_1 \geq 0, \qquad x_2 \geq 0.$$

Let x_3 and x_4 denote the slack variables for the respective functional constraints. When $c_1 = 3$, $c_2 = -2$, $b_1 = 30$, and $b_2 = 10$,

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	Right Side
Z	(0)	1	0	0	1	1	40
x_2	(1)	0	0	1	1	−2	10
x_1	(2)	0	1	0	1	−1	20

the simplex method yields the following final simplex tableau.

I **(a)** Use graphical analysis to determine the allowable range for c_1 and c_2.

(b) Use algebraic analysis to derive and verify your answers in part (*a*).

I **(c)** Use graphical analysis to determine the allowable range for b_1 and b_2.

(d) Use algebraic analysis to derive and verify your answers in part (*c*)

C **(e)** Use a software package based on the simplex method to find these allowable ranges.

I **7.2-9.** Consider Variation 5 of the Wyndor Glass Co. model (see Fig. 7.5 and Table 7.8), where the changes in the parameter values given in Table 7.5 are $\bar{c}_2 = 3$, $\bar{a}_{22} = 3$, and $\bar{a}_{32} = 4$. Use the formula $\mathbf{b}^* = \mathbf{S}^*\bar{\mathbf{b}}$ to find the allowable range for each b_i. Then interpret each allowable range graphically.

I **7.2-10.** Consider Variation 5 of the Wyndor Glass Co. model (see Fig. 7.5 and Table 7.8), where the changes in the parameter values given in Table 7.5 are $\bar{c}_2 = 3$, $\bar{a}_{22} = 3$, and $\bar{a}_{32} = 4$. Verify both algebraically and graphically that the allowable range for c_1 is $c_1 \geq \frac{9}{4}$.

7.2-11. For the problem given in Table 7.5, find the allowable range for c_2. Show your work algebraically, using the tableau given in Table 7.5. Then justify your answer from a geometric viewpoint, referring to Fig. 7.2.

7.2-12.* For the original Wyndor Glass Co. problem, use the last tableau in Table 4.8 to do the following.

(a) Find the allowable range for each b_i.

(b) Find the allowable range for c_1 and c_2.

C **(c)** Use a software package based on the simplex method to find these allowable ranges.

7.2-13. For Variation 6 of the Wyndor Glass Co. model presented in Sec. 7.2, use the last tableau in Table 7.9 to do the following.

(a) Find the allowable range for each b_i.

(b) Find the allowable range for c_1 and c_2.

C **(c)** Use a software package based on the simplex method to find these allowable ranges.

7.2-14. Consider the following problem.

$$\text{Maximize} \qquad Z = 2x_1 - x_2 + 3x_3,$$

subject to

$$\begin{aligned} x_1 + x_2 + x_3 &= 3 \\ x_1 - 2x_2 + x_3 &\geq 1 \\ 2x_2 + x_3 &\leq 2 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

Suppose that the Big M method (see Sec. 4.6) is used to obtain the initial (artificial) BF solution. Let $\bar{x}_4$ be the artificial slack variable for the first constraint, x_5 the surplus variable for the second constraint, $\bar{x}_6$ the artificial variable for the second constraint, and x_7 the slack variable for the third constraint. The corresponding final set of equations yielding the optimal solution is

$$\begin{array}{lrcrcrcrcrcrcr} (0) & Z & + & 5x_2 & + & (M+2)\bar{x}_4 & + & M\bar{x}_6 & + & x_7 & = & 8 \\ (1) & x_1 & - & x_2 & + & \bar{x}_4 & & & - & x_7 & = & 1 \\ (2) & & & 2x_2 & + & x_3 & & & + & x_7 & = & 2 \\ (3) & & & 3x_2 & + & \bar{x}_4 & + & x_5 - \bar{x}_6 & & & = & 2. \end{array}$$

Suppose that the original objective function is changed to $Z = 2x_1 + 3x_2 + 4x_3$ and that the original third constraint is changed to $2x_2 + x_3 \leq 1$. Use the sensitivity analysis procedure to revise the final set of equations (in tableau form) and convert it to proper form from Gaussian elimination for identifying and evaluating the current basic solution. Then test this solution for feasibility and for optimality. (Do not reoptimize.)

7.3-1. Consider the following problem.

$$\text{Maximize} \qquad Z = 2x_1 + 5x_2,$$

subject to

$$\begin{aligned} x_1 + 2x_2 &\leq 10 \text{ (resource 1)} \\ x_1 + 3x_2 &\leq 12 \text{ (resource 2)} \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0,$$

where Z measures the profit in dollars from the two activities.

While doing sensitivity analysis, you learn that the estimates of the unit profits are accurate only to within ±50 percent. In other words, the ranges of *likely values* for these unit profits are \$1 to \$3 for activity 1 and \$2.50 to \$7.50 for activity 2.

E* **(a)** Formulate a spreadsheet model for this problem based on the original estimates of the unit profits. Then use Solver to find an optimal solution and to generate the sensitivity report.

E* **(b)** Use the spreadsheet and Solver to check whether this optimal solution remains optimal if the unit profit for activity 1 changes from \$2 to \$1. From \$2 to \$3.

E* **(c)** Also check whether the optimal solution remains optimal if the unit profit for activity 1 still is \$2 but the unit profit for activity 2 changes from \$5 to \$2.50. From \$5 to \$7.50.

E* **(d)** Use a parameter analysis report to systematically generate the optimal solution and total profit as the unit profit of activity 1 increases in 20¢ increments from \$1 to \$3 (without changing the unit profit of activity 2). Then do the same as the unit profit of activity 2 increases in 50¢ incre-

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ments from \$2.50 to \$7.50 (without changing the unit profit of activity 1). Use these results to estimate the allowable range for the unit profit of each activity.

I **(e)** Use the Graphical Method and Sensitivity Analysis procedure in IOR Tutorial to estimate the allowable range for the unit profit of each activity.

E* **(f)** Use the sensitivity report provided by Solver to find the allowable range for the unit profit of each activity. Then use these ranges to check your results in parts (*b*–*e*).

E* **(g)** Use a two-way parameter analysis report to systematically generate the optimal solution as the unit profits of the two activities are changed simultaneously as described in part (*d*).

I **(h)** Use the Graphical Method and Sensitivity Analysis procedure in IOR Tutorial to interpret the results in part (*g*) graphically.

E* **7.3-2.** Reconsider the model given in Prob. 7.3-1. While doing sensitivity analysis, you learn that the estimates of the right-hand sides of the two functional constraints are accurate only to within ±50 percent. In other words, the ranges of *likely values* for these parameters are 5 to 15 for the first right-hand side and 6 to 18 for the second right-hand side.

(a) After solving the original spreadsheet model, determine the shadow price for the first functional constraint by increasing its right-hand side by 1 and solving again.

(b) Use a parameter analysis report to generate the optimal solution and total profit as the right-hand side of the first functional constraint is incremented by 1 from 5 to 15. Use this report to estimate the allowable range for this right-hand side, i.e., the range over which the shadow price obtained in part (*a*) is valid.

(c) Repeat part (*a*) for the second functional constraint.

(d) Repeat part (*b*) for the second functional constraint where its right-hand side is incremented by 1 from 6 to 18.

(e) Use Solver's sensitivity report to determine the shadow price for each functional constraint and the allowable range for the right-hand side of each of these constraints.

7.3-3. Consider the following problem.

Maximize $Z = x_1 + 2x_2$,

subject to

$x_1 + 3x_2 \leq 8$ (resource 1)
$x_1 + x_2 \leq 4$ (resource 2)

and

$x_1 \geq 0, \quad x_2 \geq 0,$

where Z measures the profit in dollars from the two activities and the right-hand sides are the number of units available of the respective resources.

I **(a)** Use the graphical method to solve this model.

I **(b)** Use graphical analysis to determine the shadow price for each of these resources by solving again after increasing the amount of the resource available by 1.

E* **(c)** Use the spreadsheet model and Solver instead to do parts (*a*) and (*b*).

E* **(d)** For each resource in turn, use a parameter analysis report to systematically generate the optimal solution and the total profit when the only change is that the amount of that resource available increases in increments of 1 from 4 less than the original value up to 6 more than the current value. Use these results to estimate the allowable range for the amount available of each resource.

(e) Use Solver's sensitivity report to obtain the shadow prices. Also use this report to find the range for the amount of each resource available over which the corresponding shadow price remains valid.

(f) Describe why these shadow prices are useful when management has the flexibility to change the amounts of the resources being made available.

7.3-4.* One of the products of the G.A. Tanner Company is a special kind of toy that provides an estimated unit profit of \$3. Because of a large demand for this toy, management would like to increase its production rate from the current level of 1,000 per day. However, a limited supply of two subassemblies (A and B) from vendors makes this difficult. Each toy requires two subassemblies of type A, but the vendor providing these subassemblies would only be able to increase its supply rate from the current 2,000 per day to a maximum of 3,000 per day. Each toy requires only one subassembly of type B, but the vendor providing these subassemblies would be unable to increase its supply rate above the current level of 1,000 per day. Because no other vendors currently are available to provide these subassemblies, management is considering initiating a new production process internally that would simultaneously produce an equal number of subassemblies of the two types to supplement the supply from the two vendors. It is estimated that the company's cost for producing one subassembly of each type would be \$2.50 more than the cost of purchasing these subassemblies from the two vendors. Management wants to determine both the production rate of the toy and the production rate of each pair of subassemblies (one A and one B) that would maximize the total profit.

The following table summarizes the data for the problem.

	Resource Usage per Unit of Each Activity		
	Activity		
Resource	**Produce Toys**	**Produce Subassemblies**	**Amount of Resource Available**
Subassembly A	2	–1	3,000
Subassembly B	1	–1	1,000
Unit profit	\$3	–\$2.50	

E* **(a)** Formulate and solve a spreadsheet model for this problem.

E* **(b)** Since the stated unit profits for the two activities are only estimates, management wants to know how much each of these estimates can be off before the optimal solution would change. Begin exploring this question for the first activity (producing toys) by using the spreadsheet and Solver to manually generate a table that gives the optimal solution and total profit as the unit profit for this activity increases in 50¢ increments from \$2 to \$4. What conclusion can be drawn about how much the estimate of this unit profit can

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differ in each direction from its original value of \$3 before the optimal solution would change?

E* **(c)** Repeat part (*b*) for the second activity (producing subassemblies) by generating a table as the unit profit for this activity increases in 50¢ increments from –\$3.50 to –\$1.50 (with the unit profit for the first activity fixed at \$3).

E* **(d)** Use the parameter analysis report to systematically generate all the data requested in parts (*b*) and (*c*), except use 25¢ increments instead of 50¢ increments. Use these data to refine your conclusions in parts (*b*) and (*c*).

I **(e)** Use the Graphical Method and Sensitivity Analysis procedure in IOR Tutorial to determine how much the unit profit of each activity can change in either direction (without changing the unit profit of the other activity) before the optimal solution would change. Use this information to specify the allowable range for the unit profit of each activity.

E* **(f)** Use Solver's sensitivity report to find the allowable range for the unit profit of each activity.

E* **(g)** Use a two-way parameter analysis report to systematically generate the optimal solution as the unit profits of the two activities are changed simultaneously as described in parts (*b*) and (*c*).

(h) Use the information provided by Solver's sensitivity report to describe how far the unit profits of the two activities can change simultaneously before the optimal solution might change.

E* **7.3-5.** Reconsider Prob. 7.3-4. After further negotiations with each vendor, management of the G.A. Tanner Co. has learned that either of them would be willing to consider increasing their supply of their respective subassemblies over the previously stated maxima (3,000 subassemblies of type A per day and 1,000 of type B per day) if the company would pay a small premium over the regular price for the extra subassemblies. The size of the premium for each type of subassembly remains to be negotiated. The demand for the toy being produced is sufficiently high so that 2,500 per day could be sold if the supply of subassemblies could be increased enough to support this production rate. Assume that the original estimates of unit profits given in Prob. 7.3-4 are accurate.

(a) Formulate and solve a spreadsheet model for this problem with the original maximum supply levels and the additional constraint that no more than 2,500 toys should be produced per day.

(b) Without considering the premium, use the spreadsheet and Solver to determine the shadow price for the subassembly A constraint by solving the model again after increasing the maximum supply by 1. Use this shadow price to determine the maximum premium that the company should be willing to pay for each subassembly of this type.

(c) Repeat part (*b*) for the subassembly B constraint.

(d) Estimate how much the maximum supply of subassemblies of type A could be increased before the shadow price (and the corresponding premium) found in part (*b*) would no longer be valid by using a parameter analysis report to generate the optimal solution and total profit (excluding the premium) as the maximum supply increases in increments of 100 from 3,000 to 4,000.

(e) Repeat part (*d*) for subassemblies of type B by using a parameter analysis report as the maximum supply increases in increments of 100 from 1,000 to 2,000.

(f) Use Solver's sensitivity report to determine the shadow price for each of the subassembly constraints and the allowable range for the right-hand side of each of these constraints.

E* **7.3-6.*** Consider the Union Airways problem presented in Sec. 3.4, including the data given in Table 3.19. The Excel files for Chap. 3 include a spreadsheet that shows the formulation and optimal solution for this problem. You are to use this spreadsheet and Solver to do parts (*a*) to (*g*) below.

Management is about to begin negotiations on a new contract with the union that represents the company's customer service agents. This might result in some small changes in the daily costs per agent given in Table 3.19 for the various shifts. Several possible changes listed below are being considered separately. In each case, management would like to know whether the change might result in the solution in the spreadsheet no longer being optimal. Answer this question in parts (*a*) to (*e*) by using the spreadsheet and Solver directly. If the optimal solution changes, record the new solution.

(a) The daily cost per agent for Shift 2 changes from \$160 to \$165.

(b) The daily cost per agent for Shift 4 changes from \$180 to \$170.

(c) The changes in parts (*a*) and (*b*) both occur.

(d) The daily cost per agent increases by \$4 for shifts 2, 4, and 5, but decreases by \$4 for shifts 1 and 3.

(e) The daily cost per agent increases by 2 percent for each shift.

(f) Use Solver to generate the sensitivity report for this problem. Suppose that the above changes are being considered later without having the spreadsheet model immediately available on a computer. Show in each case how the sensitivity report can be used to check whether the original optimal solution must still be optimal.

(g) For each of the five shifts in turn, use a parameter analysis report to systematically generate the optimal solution and total cost when the only change is that the daily cost per agent on that shift increases in \$3 increments from \$15 less than the current cost up to \$15 more than the current cost.

E* **7.3-7.** Reconsider the Union Airways problem and its spreadsheet model that was dealt with in Prob. 7.3-6.

Management now is considering increasing the level of service provided to customers by increasing one or more of the numbers in the rightmost column of Table 3.19 for the minimum number of agents needed in the various time periods. To guide them in making this decision, they would like to know what impact this change would have on total cost.

Use Solver to generate the sensitivity report in preparation for addressing the following questions.

(a) Which of the numbers in the rightmost column of Table 3.19 can be increased without increasing total cost? In each case, indicate how much it can be increased (if it is the only one being changed) without increasing total cost.

(b) For each of the other numbers, how much would the total cost increase per increase of 1 in the number? For each answer, indicate how much the number can be increased (if it is the only one being changed) before the answer is no longer valid.

(c) Do your answers in part (*b*) definitely remain valid if all the numbers considered in part (*b*) are simultaneously increased by one?

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(d) Do your answers in part (*b*) definitely remain valid if all 10 numbers are simultaneously increased by one?

(e) How far can all 10 numbers be simultaneously increased by the same amount before your answers in part (*b*) may no longer be valid?

7.3–8 David, LaDeana, and Lydia are the sole partners and workers in a company which produces fine clocks. David and LaDeana each are available to work a maximum of 40 hours per week at the company, while Lydia is available to work a maximum of 20 hours per week.

The company makes two different types of clocks: a grandfather clock and a wall clock. To make a clock, David (a mechanical engineer) assembles the inside mechanical parts of the clock while LaDeana (a woodworker) produces the handcarved wood casings. Lydia is responsible for taking orders and shipping the clocks. The amount of time required for each of these tasks is shown below.

	Time Required	
Task	**Grandfather Clock**	**Wall Clock**
Assemble clock mechanism	6 hours	4 hours
Carve wood casing	8 hours	4 hours
Shipping	3 hours	3 hours

Each grandfather clock built and shipped yields a profit of $300, while each wall clock yields a profit of $200.

The three partners now want to determine how many clocks of each type should be produced per week to maximize the total profit.

(a) Formulate a linear programming model in algebraic form for this problem.

I **(b)** Use the Graphical Method and Sensitivity Analysis procedure in IOR Tutorial to solve the model. Then use this procedure to check if the optimal solution would change if the unit profit for grandfather clocks is changed from $300 to $375 (with no other changes in the model). Then check if the optimal solution would change if, in addition to this change in the unit profit for grandfather clocks, the estimated unit profit for wall clocks also changes from $200 to $175.

E* **(c)** Formulate and solve this model on a spreadsheet.

E* **(d)** Use Solver to check the effect of the changes specified in part (*b*).

E* **(e)** Use a parameter analysis report to systematically generate the optimal solution and total profit as the unit profit for grandfather clocks is increased in $20 increments from $150 to $450 (with no change in the unit profit for wall clocks). Then do the same as the unit profit for wall clocks is increased in $20 increments from $50 to $350 (with no change in the unit profit for grandfather clocks). Use this information to estimate the allowable range for the unit profit of each type of clock.

E* **(f)** Use a two-way parameter analysis report to systematically generate the optimal solution as the unit profits for the two types of clocks are changed simultaneously as specified in part (*e*), except use $50 increments instead of $20 increments.

E* **(g)** For each of the three partners in turn, use Solver the effect on the optimal solution and the total profit if that partner alone were to increase the maximum number of hours available to work per week by 5 hours.

E* **(h)** Use a parameter analysis report to systematically generate the optimal solution and the total profit when the only change is that David's maximum number of hours available to work per week changes to each of the following values: 35, 37, 39, 41, 43, 45. Then do the same when the only change is that LaDeana's number changes in the same way. Then do the same when the only change is that Lydia's number changes to each of the following values: 15, 17, 19, 21, 23, 25.

E* **(i)** Generate Solver's sensitivity report and use it to determine the allowable range for the unit profit for each type of clock and the allowable range for the maximum number of hours each partner is available to work per week.

(j) To increase the total profit, the three partners have agreed that one of them will slightly increase the maximum number of hours available to work per week. The choice of which one will be based on which one would increase the total profit the most. Use the sensitivity report to make this choice. (Assume no change in the original estimates of the unit profits.)

(k) Explain why one of the shadow prices is equal to zero.

(l) Can the shadow prices in the sensitivity report be validly used to determine the effect if Lydia were to change her maximum number of hours available to work per week from 20 to 25? If so, what would be the increase in the total profit?

(m) Repeat part (*l*) if, in addition to the change for Lydia, David also were to change his maximum number of hours available to work per week from 40 to 35.

I **(n)** Use graphical analysis to verify your answer in part (*m*).

7.4-1. Reconsider the example illustrating the use of robust optimization that was presented in Sec. 7.4. Wyndor management now feels that the analysis described in this example was overly conservative for three reasons: (1) it is unlikely that the true value of a parameter will turn out to be quite near either end of its range of uncertainty shown in Table 7.10, (2) it is even more unlikely that the true values of *all* the parameters in a constraint will turn out to simultaneously lean toward the undesirable end of their ranges of uncertainty, and (3) there is a bit of latitude in each constraint to compensate for violating the constraint by a tiny bit.

Therefore, Wyndor management has asked its staff (you) to solve the model again while using ranges of uncertainty that are half as wide as those shown in Table 7.10.

(a) What is the resulting optimal solution and how much would this increase the total profit per week?

(b) If Wyndor would need to pay a penalty of $5000 per week to the distributor if the production rates fall below these new guaranteed minimum amounts, should Wyndor use these new guarantees?

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7.4-2. Consider the following problem.

$$\text{Maximize } Z = c_1x_1 + c_2x_2,$$

subject to

$$a_{11}x_1 + a_{12}x_2 \le b_1$$
$$a_{21}x_1 + a_{22}x_2 \le b_2$$

and

$$x_1 \ge 0, \quad x_2 \ge 0.$$

The estimates and ranges of uncertainty for the parameters are shown in the next table.

Parameter	Estimate	Range of Uncertainty
a_{11}	1	0.9 – 1.1
a_{12}	2	1.6 – 2.4
a_{21}	2	1.8 – 2.2
a_{22}	1	0.8 – 1.2
b_1	9	8.5 – 9.5
b_2	8	7.6 – 8.4
c_1	3	2.7 – 3.3
c_2	4	3.6 – 4.4

(a) Use the graphical method to solve this model when using the estimates of the parameters.

(b) Now use robust optimization to formulate a conservative version of this model. Use the graphical method to solve this model. Show the values of Z obtained in parts (a) and (b) and then calculate the percentage change in Z by replacing the original model by the robust optimization model.

7.4.3. Follow the instructions of Prob. 7.4-2 when considering the following problem and the information provided about its parameters in the table below.

$$\text{Minimize } Z = c_1x_1 + c_2x_2,$$

subject to the constraints shown at the top of the next column.

Parameter	Estimate	Range of Uncertainty
a_{11}	10	6 – 12
a_{12}	5	4 – 6
a_{21}	–2	–3 to –1
a_{22}	10	8 – 12
a_{31}	5	4 – 6
a_{32}	5	3 – 8
b_1	50	45 – 60
b_2	20	15 – 25
b_3	30	27 – 32
c_1	20	18 – 24
c_2	15	12 – 18

$$a_{11}x_1 + a_{12}x_2 \le b_1$$
$$a_{21}x_1 + a_{22}x_2 \le b_2$$
$$a_{31}x_1 + a_{32}x_2 \le b_3$$

and

$$x_1 \ge 0, \quad x_2 \ge 0.$$

C **7.4-4.** Consider the following problem.

$$\text{Maximize } Z = 5x_1 + c_2x_2 + c_3x_3,$$

subject to

$$a_{11}x_1 - 3x_2 + 2x_3 \le b_1$$
$$3x_1 + a_{22}x_2 + x_3 \ge b_2$$
$$2x_1 - 4x_2 + a_{33}x_3 \le 20$$

and

$$x_1 \ge 0, \quad x_2 \ge 0, \quad x_3 \ge 0.$$

The estimates and ranges of uncertainty for the uncertain parameters are shown in the next table.

Parameter	Estimate	Range of Uncertainty
a_{11}	4	3.6 – 4.4
a_{22}	−1	−1.4 to −0.6
a_{33}	3	2.5 – 3.5
b_1	30	27 – 33
b_2	20	19 – 22
c_2	−8	−9 to −7
c_3	4	3 – 5

(a) Solve this model when using the estimates of the parameters.

(b) Now use robust optimization to formulate a conservative version of this model. Solve this model. Show the values of Z obtained in parts (a) and (b) and then calculate the percentage decrease in Z by replacing the original model by the robust optimization model.

7.5-1. Reconsider the example illustrating the use of chance constraints that was presented in Sec. 7.5. The concern is that there is some uncertainty about how much production time will be available for Wyndorís two new products when their production begins in the three plants a little later. Table 7.11 shows the initial estimates of the mean and standard deviation of the available production time per week in each of the three plants.

Suppose now that a more careful investigation of these available production times has considerably narrowed down the range of what these times might turn out to be with any significant likelihood. In particular, the means in Table 7.11 remain the same but the standard deviations have been cut in half. However, to add more insurance that the original constraints

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still will hold when production begins, the value of α has been increased to $\alpha = 0.99$. It is still assumed that the available production time in each plant has a normal distribution.

(a) Use probability expressions to write the three chance constraints. Then show the deterministic equivalents of these chance constraints.

(b) Solve the resulting linear programming model. How much total profit per week would this solution provide to Wyndor? Compare this total profit per week to what was obtained for the example in Sec. 7.5. What is the increase in total profit per week that was enabled by the more careful investigation that cut the standard deviations in half?

7.5-2. Consider the following constraint whose right-hand side b is assumed to have a normal distribution with a mean of 100 and some standard deviation σ.

$$30x_1 + 20x_2 \leq b$$

A quick investigation of the possible spread of the random variable b has led to the estimate that $\sigma = 10$. However, a subsequent more careful investigation has greatly narrowed down this spread, which has led to the refined estimate that $\sigma = 2$. After choosing a minimum acceptable probability that the constraint will hold (denoted by α) this constraint will be treated as a chance constraint.

(a) Use a probability expression to write the resulting chance constraint. Then write its deterministic equivalent in terms of σ and K_α.

(b) Prepare a table that compares the value of the right-hand side of this deterministic equivalent for $\sigma = 10$ and $\sigma = 2$ when using $\alpha = 0.9$, 0.95, 0.975, 0.99, and 0.99865.

7.5-3. Suppose that a linear programming problem has 20 functional constraints in inequality form such that their right-hand sides (the b_i) are uncertain parameters, so chance constraints with some α are introduced in place of these constraints. After next substituting the deterministic equivalents of these chance constraints and solving the resulting new linear programming model, its optimal solution is found to satisfy 10 of these deterministic equivalents with equality whereas there is some slack in the other 10 deterministic equivalents. Answer the following questions under the assumption that the 20 uncertain b_i have mutually independent normal distributions.

(a) When choosing $\alpha = 0.95$, what are the lower bound and upper bound on the probability that *all* of these 20 original constraints will turn out to be satisfied by the optimal solution for the new linear programming problem so this solution actually will be feasible for the original problem.

(b) Now repeat part (a) with $\alpha = 0.99$.

(c) Suppose that all 20 of these functional constraints are considered to be *hard constraints*, i.e., constraints that *must* be satisfied if at all possible. Therefore, the decision maker desires to use a value of α that will guarantee a probability of at least 0.95 that the optimal solution for the new linear programming problem actually will turn out to be feasible for the original problem. Use trial and error to find the smallest value of α (to three significant digits) that will provide the decision maker with the desired guarantee.

7.5.4 Consider the following problem.

$$\text{Maximize} \quad Z = 20x_1 + 30x_2 + 25x_3,$$

subject to

$$\begin{aligned} 3x_1 + 2x_2 + x_3 &\leq b_1 \\ 2x_1 + 4x_2 + 2x_3 &\leq b_1 \\ x_1 + 3x_2 + 5x_3 &\leq b_3 \end{aligned}$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0,$$

where b_1, b_2, and b_3 are uncertain parameters that have mutually independent normal distributions. The mean and standard deviation of these parameters are (90, 3), (150, 6), and (180, 9), respectively.

(a) The proposal has been made to use the solution, $(x_1, x_2, x_3) = (7, 22, 19)$. What are the probabilities that the respective functional constraints will be satisfied by this solution?

C **(b)** Formulate chance constraints for these three functional constraints where $\alpha = 0.975$ for the first constraint, $\alpha = 0.95$ for the second constraint, and $\alpha = 0.90$ for the third constraint. Then determine the deterministic equivalents of the three chance constraints and solve for the optimal solution for the resulting linear programming model.

(c) Calculate the probability that the optimal solution for this new linear programming model will turn out to be feasible for the original problem.

C **7.6-1.** Reconsider the example illustrating the use of stochastic programming with recourse that was presented in Sec. 7.6. Wyndor management now has obtained additional information about the rumor that a competitor is planning to produce and market a special new product that would compete directly with Wyndor's product 2. This information suggests that it is less likely that the rumor is true than was originally thought. Therefore, the estimate of the probability that the rumor is true has been reduced to 0.5.

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Formulate the revised stochastic programming model and solve for its optimal solution. Then describe the corresponding optimal plan in words.

C 7.6-2. The situation is the same as described in Prob. 7.6-1 except that Wyndor management does not consider the additional information about the rumor to be reliable. Therefore, they havenít yet decided whether their best estimate of the probability that the rumor is true should be 0.5 or 0.75 or something in between. Consequently, they have asked you to find the break-even point for this probability below which the optimal plan presented in Sec. 7.6 will no longer be optimal. Use trial and error to find this break-even point (rounded up to two decimal points). What is the new optimal plan if the probability is a little less than this break-even point?

C 7.6-3. The Royal Cola Company is considering developing a special new carbonated drink to add to its standard product line of drinks for a couple years or so (after which it probably would be replaced by another special drink). However, it is unclear whether the new drink would be profitable, so analysis is needed to determine whether to go ahead with the development of the drink. If so, once the development is completed, the new drink would be marketed in a small regional test market to assess how popular the drink would become. If the test market suggests that the drink should become profitable, it then would be marketed nationally.

Here are the relevant data. The cost of developing the drink and then arranging to test it in the test market is estimated to be \$40 million. A total budget of \$100 million has been allocated to advertising the drink in both the test market and nationally (if it goes national). A minimum of \$5 million is needed for advertising in the test market and the maximum allowed for this purpose would be \$10 million, which would leave between \$90 million and \$95 million for national advertising. To simplify the analysis, sales in either the test market or nationally is assumed to be proportional to the level of advertising there (while recognizing that the rate of additional sales would fall off after the amount of advertising reaches a saturation level). Excluding the fixed cost of \$40 million, the net profit in the test market is expected to be half the level of advertising.

To further simplify the analysis, the outcome of testing the drink in the test market would fall into just three categories: (1) very favorable, (2) barely favorable, (3) unfavorable. The probabilities of these outcomes are estimated to be 0.25, 0.25, and 0.50, respectively. If the outcome were *very favorable*, the net profit after going national would be expected to be about twice the level of advertising. The corresponding net profit if the outcome were *barely favorable* would be about 0.2 times the level of advertising. If the outcome were *unfavorable*, the drink would be dropped and so would not be marketed nationally.

Use stochastic programming with recourse to formulate a model for this problem. Assuming the company should go ahead with developing the drink, solve the model to determine how much advertising should be done in the test market and then how much advertising should be done nationally (if any) under each of the three possible outcomes in the test market. Finally, calculate the expected value (in the statistical sense) of the total net profit from the drink, including the fixed cost if the company goes ahead with developing the drink, where the company should indeed go ahead only if the expected total net profit is positive.

C 7.6-4. Consider the following problem.

$$\text{Minimize} \quad Z = 5x_1 + c_2x_2,$$

subject to

$$3x_1 + a_{12}x_2 \geq 60$$
$$2x_1 + a_{22}x_2 \geq 60$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0,$$

where x_1 represents the level of activity 1 and x_2 represents the level of activity 2. The values of c_2, a_{12}, and a_{22} have not been determined yet. Only activity 1 needs to be undertaken soon whereas activity 2 will be initiated somewhat later. There are different scenarios that could unfold between now and the time activity 2 is undertaken that would lead to different values for c_2, a_{12}, and a_{22}. Therefore, the goal is to use all of this information to choose a value for x_1 now and to simultaneously determine a plan for choosing a value of x_2 later after seeing which scenario has occurred.

Three scenarios are considered plausible possibilities. They are listed below, along with the values of c_2, a_{12}, and a_{22} that would result from each one:

Scenario 1: $c_2 = 4$, $a_{12} = 2$, and $a_{22} = 3$
Scenario 2: $c_2 = 6$, $a_{12} = 3$, and $a_{22} = 4$
Scenario 3: $c_2 = 3$, $a_{12} = 2$, and $a_{22} = 1$

These three scenarios are considered equally likely.

Use stochastic programming with recourse to formulate the appropriate model for this problem and then to solve for the optimal plan.

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